

Occupant Protection

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201 (1) In this section,

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head impact area

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head impact area means the non-glazed surfaces of the interior of the vehicle that are capable of being contacted statically by the head form of a measuring device in accordance with the following procedure or its graphic equivalent:

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(a) at each designated seating position, by placing the pivot point of the measuring device,

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(i) for seats that are adjustable fore and aft, at

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(A) the seating reference point, and

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(B) a point 130 mm (five inches) horizontally forward of the seating reference point displaced vertically 20 mm (0.75 inches) or a distance equal to the rise that results from a 130 mm (five inches) forward adjustment of the seat, and

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- (ii) for seats that are not adjustable fore and aft, at the seating reference point,
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- (b) with the pivot point to top-of-head dimension at each adjustment allowed for the measuring device by the interior dimension of the vehicle, by determining all contact points above the lower windshield glass line and forward of the seating reference point, and
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- (c) beginning with the head form of the measuring device at each contact point referred to in paragraph (b) and with the measuring device in a vertical position if no contact point exists for a particular adjusted length, by pivoting the measuring device forward and downward through all arcs in vertical planes to 90° each side of the longitudinal plane through the seating reference point until the head form contacts an interior surface or until it is tangent to a horizontal plane 25 mm (1 inch) above the seating reference point, whichever occurs first; (zone d'impact de la tête)
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interior compartment door

interior compartment door means any door in the interior of a vehicle installed by the manufacturer as a cover for storage space normally used for personal effects; (porte de compartiment intérieur)

measuring device

measuring device means a device having a spherical head form with a diameter of 165 mm (6.5 inches) attached to a pivot, with the distance between the pivot point and the top of the head form being adjustable at any point between 740 mm (29 inches) and 840 mm (33 inches); (appareil de mesure)

pelvic impact area

pelvic impact area means that area of the door or body side panel adjacent to any outboard designated seating position that is bounded by:

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- (a) horizontal planes 180 mm (seven inches) above and 100 mm (four inches) below the seating reference point, and

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(b) vertical transverse planes 200 mm (eight inches) forward and 50 mm (two inches) rearward of the seating reference point. (zone d'impact pelvienne)

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(2) This section applies to passenger cars and to multi-purpose passenger vehicles, trucks and buses with a GVWR of 4 536 kg (10,000 pounds) or less.

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(3) Subject to subsection (3.2), the instrument panel of a vehicle shall be constructed so that the deceleration of a spherical head form with a weight of 6,8 kg (15 pounds) and a diameter of 165 mm (6.5 inches) does not exceed 80 g continuously for more than 3 ms when the area of the instrument panel that is within the head impact area is impacted in accordance with the test procedure referred to in subsection (3.1) at a relative velocity of

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(a) 19.3 km/h, in the case of a vehicle that is equipped, at the right front outboard designated seating position, with an air bag and a Type 2 manual seat belt assembly that conform to section 208; and

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(b) 24 km/h (15 mph), in the case of any other vehicle.

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(3.1) The test of the instrument panel shall be performed in accordance with SAE Recommended Practice, "Motor Vehicle Instrument Panel Laboratory Impact Test Procedure — Head Area — SAE J921b" (November 1971) and using the instrumentation whose performance is specified in SAE Recommended Practice, "Instrumentation For Impact Tests — SAE J211" (October 1988) except that

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- (a) the origin of the line tangent to the surface of the instrument panel shall be a point on a transverse horizontal line through a point 130 mm (5 inches) horizontally forward of the seating reference point of the front outboard designated seating position, displaced vertically 19 mm (0.75 inches) or a distance equal to the rise in the seating reference point that results from a 130 mm (5 inch) forward adjustment of the seat; and

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- (b) the impact shall be directed in a vertical plane parallel to the longitudinal axis of the vehicle or in a plane normal to the surface at the point of contact.

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(3.2) Subsection (3) does not apply to

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- (a) console assemblies;

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- (b) areas less than 130 mm (5 inches) from the junction of the instrument panel and the inside of the side structure of the body;

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- (c) areas closer to the junction of the windshield and the instrument panel than those areas that are capable of being contacted by the head form with the windshield in place;

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(d) areas outboard of all points on the line of contact of the head form with the instrument panel when the head form is moved along the inboard side of a vertical longitudinal plane tangent to the inboard edge of the steering wheel; or

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(e) areas below any point at which a vertical line is tangent to the rearmost surface of the instrument panel.

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(4) Subject to subsection (4.1), the seats of a vehicle shall be constructed so that the deceleration of a spherical head form with a weight of 6,8 kg (15 pounds) and a diameter of 165 mm (6.5 inches) does not exceed 80 g continuously for more than 3 ms when the area of a seatback that is within the head impact area is impacted, at a relative velocity of 24 km/h (15 miles per hour), in accordance with SAE Recommended Practice, “Motor Vehicle Seatback Assembly Laboratory Impact Test Procedure — Head Area — SAE J346” (June 1971) using the instrumentation whose performance is specified in SAE Recommended Practice, “Instrumentation For Impact Tests — SAE J211” (October 1988), except that

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(a) the origin of the line tangent to the uppermost component of the seatback frame shall be a point on a transverse horizontal line through the seating reference point of the appropriate rear outboard designated seating position, with adjustable forward seats in their rearmost design driving position and reclinable forward seatbacks in their normal design position;

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(b) the impact shall be directed in a vertical plane parallel to the longitudinal axis of the vehicle or in a plane normal to the surface at the point of contact;

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(c) in the case of

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(i) a seat without head restraints, each test shall be performed

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(A) in respect of each individual split seatback or bucket seat seatback, between two points each situated 100 mm (4 inches) outboard of its centreline, and

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(B) in respect of each bench seat seatback, between two points each situated 100 mm (4 inches) outboard of the centreline of each outboard designated seating position, and

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(ii) a seat with head restraints, each test shall be conducted at a point on the centreline of each head restraint with the head restraint in place at its lowest adjusted position; and

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(d) in the case of a type of seat that is installed in more than one body style, tests conducted at the foremost and rearmost positions identified by the application of paragraph (a) shall be deemed to be conclusive for all intermediate positions.

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(4.1) Subsection (4) does not apply in respect of

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(a) seats installed in a school bus that complies with section 222 when tested in accordance with Motor Vehicle Safety Test Methods, section 222, "School Bus Passenger Seating and Crash Protection" (December 19, 1983);

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(b) the rearmost seats;

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(c) side-facing seats;

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(d) back-to-back seats;

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(e) folding auxiliary jump seats; and

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(f) temporary seats.

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(5) Subject to subsection (7), every interior compartment door located in an instrument panel, console, seatback or side panel adjacent to an outboard designated seating position shall remain closed when

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(a) the interior compartment door latch system is tested by being subjected to a force equal to an inertia load of 10 g in a horizontal transverse direction and an inertia load of 10 g in a vertical direction, in accordance with SAE Recommended Practice, "Passenger Car Side Door Latch Systems — SAE J839" (July 1982);

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(b) one of the following tests is conducted, namely,

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- (i) the vehicle is tested by being impacted perpendicularly into a fixed collision barrier at a forward longitudinal velocity of 48 km/h (30 miles per hour), or

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- (ii) the interior compartment door latch system is tested by being subjected to a force equal to a horizontal inertia load of 30 g in a longitudinal direction, in accordance with SAE Recommended Practice, "Passenger Car Side Door Latch Systems — SAE J839" (July 1982); and

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- (c) the instrument panel or seatback is tested in accordance with subsection (3) or (4), as the case may be.

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- (6) [Repealed, SOR/92-689, s. 1]

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- (7) All interior compartment door assemblies that have a locking device shall be tested with the locking device in an unlocked position.

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- (8) A sun visor shall be provided at each front outboard designated seating position on every vehicle and shall be constructed of, or covered with, energy absorbing material and so mounted that each mounting shall not have a rigid material edge with a radius of less than 3 mm (0.125 inch) that is statically contactable by a spherical head form with a diameter of 165 mm (6.5 inches).

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(9) Each arm-rest installed on a vehicle, other than an arm-rest referred to in subsection (10), shall meet at least one of the following performance requirements:

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 - (a) be constructed of energy absorbing material that collapses laterally at least 50 mm (2 inches) without permitting contact with any underlying rigid material;
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 - (b) be constructed
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 - (i) of energy absorbing material that collapses to within 32 mm (1.25 inches) of a rigid test panel surface without permitting contact with any rigid material, and
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 - (ii) so that the vertical height of any rigid material between 13 mm (0.5 inches) and 32 mm (1.25 inches) from the panel surface is not less than 25 mm (1 inch); and
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 - (c) when measured vertically in side elevation, provide at least 50 mm (2 inches) of coverage within the pelvic impact area along not less than 50 mm (2 inches) of its length.
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- (10) Each arm-rest that folds into the seat back or between two seat backs of a vehicle shall

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- (a) comply with subsection (9); or

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- (b) be constructed of or covered with energy absorbing material.

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